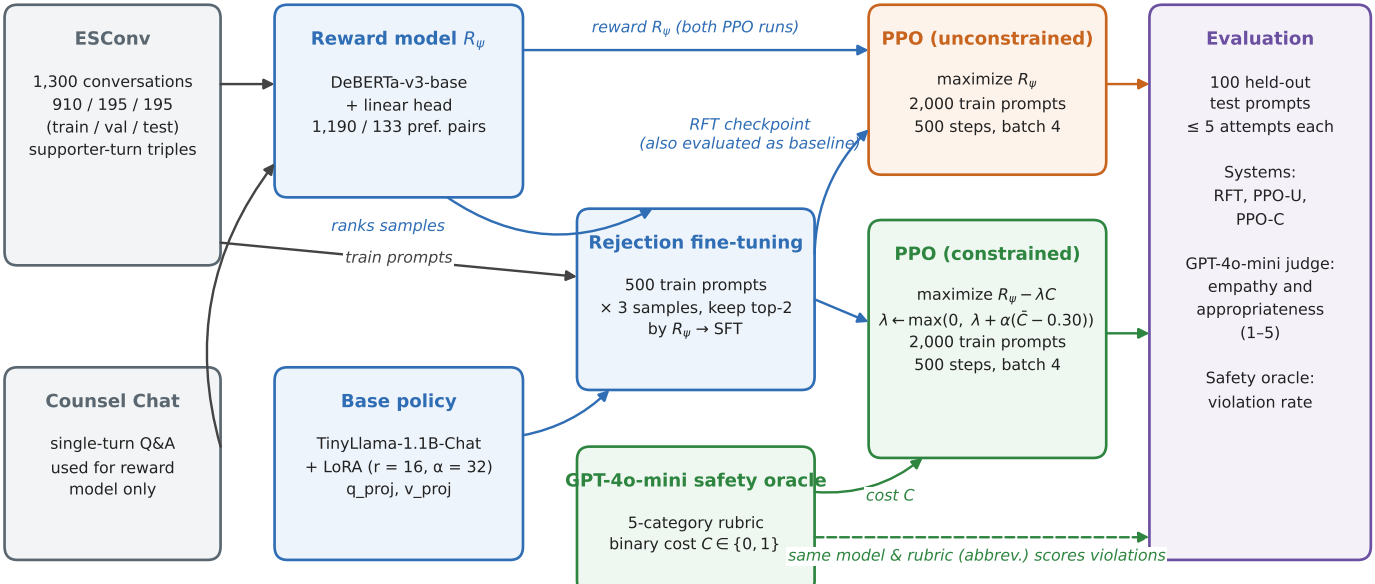
ESConv
1,300 conversations
910 / 195 / 195
(train / val / test)
supporter-turn triples
Reward model R_ψ
DeBERTa-v3-base
+ linear head
1,190 / 133 pref. pairs
reward R_ψ (both PPO runs)
PPO (unconstrained)
maximize R_ψ
2,000 train prompts
500 steps, batch 4
Evaluation
100 held-out
test prompts
≤ 5 attempts each
Systems:
RFT, PPO-U,
PPO-C
GPT-4o-mini judge:
empathy and
appropriateness
(1–5)
Safety oracle:
violation rate
RFT checkpoint
(also evaluated as baseline)
ranks samples
train prompts
Rejection fine-tuning
500 train prompts
× 3 samples, keep top-2
by R_ψ → SFT
PPO (constrained)
maximize $R_\psi - \lambda C$
$\lambda \leftarrow \max(0,\ \lambda + \alpha(\bar{C} - 0.30))$
2,000 train prompts
500 steps, batch 4
Counsel Chat
single-turn Q&A
used for reward
model only
Base policy
TinyLlama-1.1B-Chat
+ LoRA (r = 16, α = 32)
q_proj, v_proj
GPT-4o-mini safety oracle
5-category rubric
binary cost $C \in \{0, 1\}$
cost C
same model & rubric (abbrev.) scores violations